

Procedure	Stages	Cpl	Loc	Blk	Op	Collective		Blocked resources
						C sq	S/W	and remarks
On an origin process with Fence Synchronization								
MPI_WIN_FENCE	i-----		nl			C sq	W1	
MPI_PUT, MPI_GET, MPI_ACCUMULATE, ...	i-s----	ic	l	nb	nb-op	-		buffer 14) 21)
MPI_WIN_FENCE	----c-f	c+f	l			C sq	W1	
On a target process								
MPI_WIN_CREATE/_ALLOCATE/_..._SHARED	i-----		nl			C sq	W1	buffer address
MPI_WIN_FENCE	--s----	ic	l			C sq	W1	buffer address+content
MPI_WIN_FENCE	----c--	c	nl			C sq	W1	buffer address
MPI_WIN_FREE	-----f	f	nl			C sq	W1	
On an origin process with General Active Target Synchronization								
MPI_WIN_START	i-----		nl			-		21)
MPI_PUT, MPI_GET, MPI_ACCUMULATE, ...	i-s----	ic	l	nb	nb-op	-		buffer 14) 21)
MPI_WIN_COMPLETE	----c-f	c+f	l			-		21)
On a target process								
MPI_WIN_CREATE/_ALLOCATE/_..._SHARED	i-----		nl			C sq	W1	buffer address
MPI_WIN_POST	--s----	ic	l			-		buffer address+content
MPI_WIN_WAIT	----c--	c	nl			-		buffer address
MPI_WIN_FREE	-----f	f	nl			C sq	W1	
On an origin process with Lock/Unlock Synchronization								
MPI_WIN_LOCK and others	i-----		nl			-		21)
MPI_PUT, MPI_GET, MPI_ACCUMULATE, ...	i-s----	ic	l	nb	nb-op	-		buffer 14) 21)
MPI_WIN_UNLOCK and others	----c-f	c+f	l			-		21)
On a target process								
MPI_WIN_CREATE/_ALLOCATE/_..._SHARED	i-----		nl			C sq	W1	buffer address
MPI_WIN_FREE	-----f	f	nl			C sq	W1	
On an origin process with Lock/Unlock Synchronization								
MPI_WIN_LOCK and others	i-----		nl			-		21)
MPI_RPUT, MPI_RGET, MPI_RACCUMULATE, ...	i-s----	ic	l	nb	nb-op	-		buffer 14) 21)
MPI_WAIT	----c-f	c+f	l			-		21)
MPI_WIN_UNLOCK and others	-----		l			-		21)
On a target process								
MPI_WIN_CREATE/_ALLOCATE/_..._SHARED	i-----		nl			C sq	W1	buffer address
MPI_WIN_FREE	-----f	f	nl			C sq	W1	